S2 Appendix
Conservative probability calculation

Since we use p≤0.05 as the condition for significance for each individual time series, for each time series there is a 0.05 chance that it will be significant. The probability for significance in a number of time series then follows a binomial distribution, for which the probability mass function is:

$$\mathrm{Pr}_b(k; n, p) = \binom{n}{k} p^k (1-p)^{n-k} \tag{1}$$

This is the probability that the event with probability p happens k times out of the n time series.

Out of the 1197 AMOC time series analysed in this study, 105 show a significant increase in a CSD indicator. If we regard both the three CSD indicators and the three different AMOC strengths and fingerprints as independent, the probability of this happening is $\mathrm{Pr}_b(105; 1197, 0.05) = 1.55e - 08$. However, neither the CSD indicators nor the AMOC strengths can be viewed as independent (see Fig 6d in the main text, and Figs. E, F and G in S1 Figure). We thus have nine sets of 133 time series, each with their own number of significant increases. For a set of 133 time series with m significant increases, we can use the binomial distribution to calculate the probability p' of m significances out of 133 as if the other sets didn't exist: $p' = \mathrm{Pr}_b(m; 133, 0.05)$. In reality, the probability is likely slightly higher than p' because we have nine sets of time series, but these sets are dependent to an unknown extent, so we cannot view the nine sets of time series as multiple experiments. The probability of the result for each strength-indicator or index-indicator combination viewed alone are shown in Table C.

We are also interested in the likelihood of the result for individual models. A model like CESM2 with 17 out of 90 significant time series looks like it would be unlikely, but it is important to quantify this to some extent. Here again we consider the CSD indicators and AMOC strengths and index separately, as there is no way to quantify their dependence. For a given CSD indicator and AMOC strength we can then first calculate the probability for the model result considered alone. For model and indicator-strength combination i this is $p'_i = \mathrm{Pr}_b(l_i; n_i, 0.05)$, where l_i is the number of significant increases out of n_i time series. However, as noted before, an event that would be unlikely if we only had one model could be likely to happen at least once for 27 models. We thus need to consider how likely it is to have at least one event of probability p'_i out of 27 instances. We can do this here because for a given CSD indicator and AMOC strength we assume that the model results are independent. This is again the binomial distribution from equation 1, but now with $n = 27$ and $p = p'_i$. We are now interested in the likelihood of at least one occurrence of a p'_i probability event, which is: $\mathrm{Pr}_b(k \geq 1; 27, p'_i) = 1 - \mathrm{Pr}_b(0; 27, p'_i)$. So combining this with the equation for p'_i, we get:

$$\mathrm{Pr}_i(l_i) = 1 - \mathrm{Pr}_b(0; 27, \mathrm{Pr}_b(l_i; n_i, 0.05)) \tag{2}$$

This is the probability that out of 27 models, we have at least one set with a probability p_i' event (which for model i is l_i out of n_i significant time series). Of course since the value of l_i, n_i and p_i' will be different for the models, this is a conservative calculation that applies only to each model individually - if we have multiple models with low $\mathrm{Pr}_i(l_i)$ the result in total is even less likely. The model indicator-strength combinations for which $\mathrm{Pr}_i(l_i)$ is less than 0.05 are shown in Table B.

These calculations assume that the CMIP6 models are completely independent from each other. This is not strictly true, and in particular some dependence in the AMOC time series may arise from multiple models using the same ocean component. However, in our results there is no correlation between the ensemble members with lower p-values and the ocean component (Fig M in S1 Figure). In addition, the four models in Table B all use different ocean components. We can therefore assume that for our purposes the CMIP6 models are independent.

	S26.5	**S35**	**ASSTI**	**All three**
At least one	28	32	13	73
One	17	21	9	47
Two	10	6	4	20
Three	1	5	0	6

Table A: Number of time series per streamfunction or SST Index with significant increases in the CSD indicators. Columns are the streamfunction strength latitude, AMOC SST Index and all three. Rows are the number of CSD indicators with a significant increase. So for example, the first cell from the top left counts the number of S26.5 timeseries that have at least one CSD indicator with a significant increase.

Model	λ **S35**	**AR1 S35**	**AR1 S26.5**
CanESM5	2.57e-02		
CESM2	2.57e-02		2.57e-02
HadGEM3-GC31-LL		1.27e-02	
MIROC6		2.57e-02	

Table B: Probability of outcome amongst all models, for models where for a given CSD indicator and AMOC time series the outcome in CMIP6 is statistically unlikely to occur amongst 27 models (see methods).

	S26.5	**S35**	**ASSTI**
λ	4.17e-03	8.31e-05	1.04e-01
AR1	2.45e-04	6.76e-04	1.52e-01
Variance	9.19e-02	4.17e-03	1.59e-01

Table C: Probability that a given set of time series has m_i out of 133 significant increases, where m_i is the number of significant increases in the indicator-index combination. This is calculated without considering the rest of the sets of time series.